International Physics Competition

Qualification Round 2026

Solution Document

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Md Mokhdum Amin Fahim

Notre Dame College, Dhaka

Bangladesh

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Date of Submission : 1 November, 2026

Number of Pages : 4

Problem A : Size of Moon's Shadow

Given,

- Full eclipse began at $t_1 = 20:31:31$
- Full eclipse ended at $t_2 = 20:32:32$
- Moon's shadow's speed $v = 0.75\,\mathrm{km/s}$

$\therefore$ The time interval $\Delta t = t_2 - t_1 = 00:01:01 = 61\,\mathrm{s}$
$\therefore$ The distance the Moon's shadow traveled,
$s = v\Delta t = 0.75\,\mathrm{km/s} \cdot 61\,\mathrm{s} = 45.75\,\mathrm{km}$

Answer

The width of the Moon's shadow is $45.75\,\mathrm{km}$

Problem B : Floating Box

Given,

- Density of box $\rho_{box} = 650\,\mathrm{kg/m^3}$
- Density of water $\rho_{water} = 1000\,\mathrm{kg/m^3}$
- The box has outer length $a = 25\,\mathrm{cm}$, width $b = 12\,\mathrm{cm}$ and hight $c = 18\,\mathrm{cm}$
- The thickness of wall is $d = 1.0\,\mathrm{cm}$

Two forces are acting on the box - wight and buoyancy. The net force is $\sum F = 0$
$\therefore$ The volume of the box,
$V = abc - (a-2d)(b-2d)(c-2d) = 5400\,\mathrm{cm^3} - 3680\,\mathrm{cm^3} = 1720\,\mathrm{cm^3}$
$\therefore$ The weight $W = mg = \rho_{box} V g$
We think that hight h of the box is beneath the water.
$\therefore$ Volume of box under water $V_w = abh$
According to Archimedes law,
$\therefore$ The buoyant force $F_b = \rho_{water} V_w g = \rho_{water} abhg$
$\therefore \sum F = W - F_b = 0 \implies F_b = W \implies \rho_{water} abhg = \rho_{box} V g$

$$\therefore h = \frac{\rho_{box} V}{\rho_{water} ab} = \frac{650\,\mathrm{kg/m^3} \cdot 1720\,\mathrm{cm^3}}{1000\,\mathrm{kg/m^3} \cdot 25\,\mathrm{cm} \cdot 12\,\mathrm{cm}} \approx 3.73\,\mathrm{cm}$$

$\therefore$ The top of the box above the water surface is $h_t = c - h \approx 14.3\,\mathrm{cm}$

Answer

The top of the box is about $14.3\,\mathrm{cm}$ high above the water surface.

Problem C : Echo Measure

Given,

- The air temperature is $20\,^\circ\mathrm{C}$
- The student hears an echo $t = 1.24\,\mathrm{s}$
- For sound in air, use the approximate relation,

$$v = \left(331 + 0.60\frac{\mathrm{T}}{^\circ\mathrm{C}}\right)\frac{\mathrm{m}}{\mathrm{s}}$$

(a)

$\therefore$ The speed of sound,

$$v = \left(331 + 0.60\frac{20\,^\circ\mathrm{C}}{^\circ\mathrm{C}}\right)\frac{\mathrm{m}}{\mathrm{s}} = 343\,\mathrm{m/s}$$

(b)

We suppose that the cliff is distance d away.
$\therefore$ The distance,

$$d = \frac{1}{2}vt = \frac{1}{2}\cdot 343\,\mathrm{m/s}\cdot 1.24\,\mathrm{s} \approx 213\,\mathrm{m}$$

(c)

Given, the new temperature $T = 32\,^\circ\mathrm{C}$
$\therefore$ The new speed of sound,

$$v_w = \left(331 + 0.60\frac{32\,^\circ\mathrm{C}}{^\circ\mathrm{C}}\right)\frac{\mathrm{m}}{\mathrm{s}} = 350.2\,\mathrm{m/s}$$

$\therefore$ The echo retures in time,

$$t_w = \frac{2d}{v_w} = \frac{2\cdot 213\,\mathrm{m}}{350.2\,\mathrm{m/s}} \approx 1.22\,\mathrm{s}$$

$\therefore$ The echo returns earlier,

$$\Delta t = t_w - t = 1.24\,\mathrm{s} - 1.22\,\mathrm{s} = 0.02\,\mathrm{s}$$

Answer

(a) 343 m/s ; (b) 213 m ; (c) 0.02 s

Problem D : Garden Pipe Split

Given,

- Splitter gauge pressure $\Delta p = 180$ kPa
- Both branches have length L and cross-sectional area A.
- $\Delta p = kLv^2$
- $k_2(\lambda) = k_1(1 + 3\lambda^2)$

Substituting $v = Q/A$,

$$\Delta p = \left(\frac{kL}{A^2}\right) Q^2$$

As $\Delta p_1 = \Delta p_2$,

$$\left(\frac{k_1(1+3\lambda^2)L}{A^2}\right) Q_2^2 = \left(\frac{k_1 L}{A^2}\right) Q_1^2$$

$$Q_2 = \frac{Q_1}{\sqrt{1+3\lambda^2}}$$

By using conservation of mass at the splitter, the total flow rate Q_{tot} is,

$$Q_{\text{tot}}(\lambda) = Q_1 + Q_2 = Q_1 + \frac{Q_1}{\sqrt{1+3\lambda^2}} = Q_1\left(1 + \frac{1}{\sqrt{1+3\lambda^2}}\right)$$

$$Q_{\text{tot}}(\lambda) = Q_1\left(1 + \frac{1}{\sqrt{1+3\lambda^2}}\right)$$

Which is the required expression for the total flow rate through the splitter.

Answer

It is shown that : $Q_{\text{tot}}(\lambda) = Q_1\left(1 + \frac{1}{\sqrt{1+3\lambda^2}}\right)$

Problem E : Lightning Rod

During thunderstorm, charge separation creates a voltage V between the cloud and the ground, forming an electric field,

$$E = -\nabla V$$

The charge density σ accumulates densely on sharp points ,as like rod's tip, because :

$$\sigma \propto 1/R$$

Here,
R is the radius of curvature of the rod's tip . This high charge concentration increases the local electric field, as :

$$E = \sigma/\varepsilon_0$$

The ionized air triggers a controlled electrical discharge path before the rest of the structure can. At the end, according to Ohm's law,

$$I = V/R$$

The lightning current flows safely through the thick conductor directly into the Earth. Thus, our buildings remain safe due to these rods.

Answer

The rod concentrates charge at its tip to increase the local electric field which causes the lightning to strike the rod and ground through wire.